Siemens-Gamesa report

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Part A.1 - Introduction, existing product, market

Siemens-Gamesa is a wind turbine manufacturer with the largest presence within the UK by quantity of employees (2,925), and has recently become involved[1] in one of the largest offshore wind farms, East Anglia[2][3], close to their blade manufacturing facility in Hull. Their largest wind turbine on offer spans a 236m diameter[4].

Regarding the UK's trade deficit, oil is the 2nd largest import (£43.7b)[5], approximately 44% of all energy used being imported [6]. Wind power provides the largest renewable proportion of electricity generation, with offshore wind now dominating onshore installations. The UK's largest use of natural gas is for electricity generation.

The world remains largely dependent on fossil fuels; a strong correlation exists between fossil fuels use and energy imports. Presumably wind occurs anywhere oceans exist, and therefore the offshore market is significant.

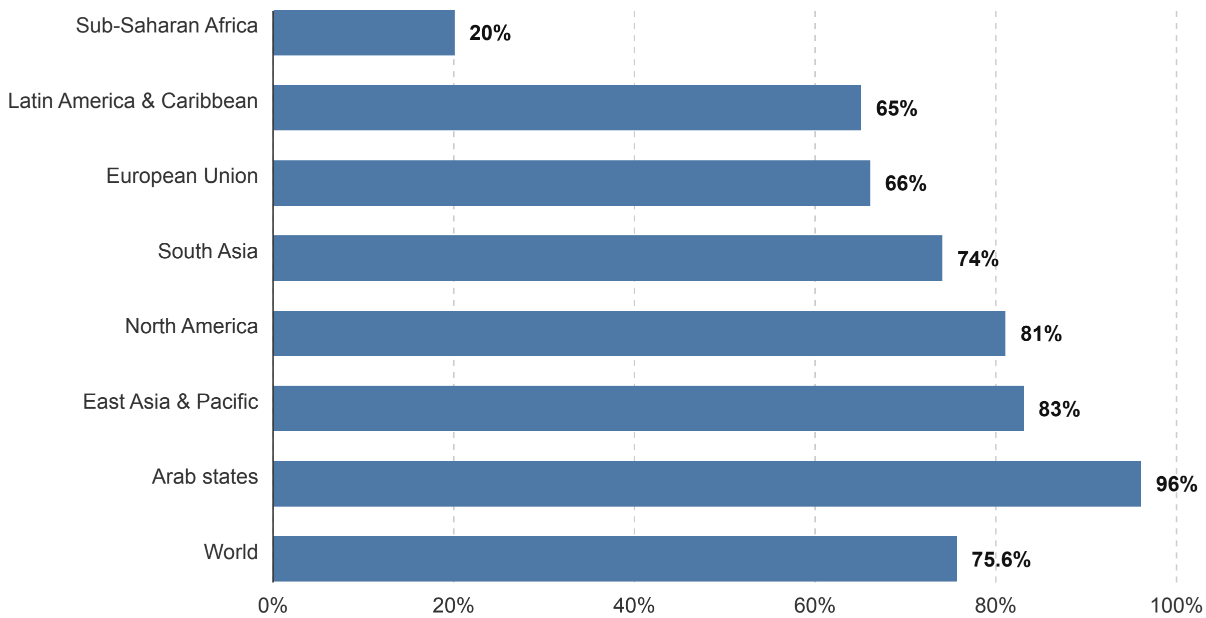

Figure 1: Bar chart to show fossil fuels as percentage of total energy use.[7]

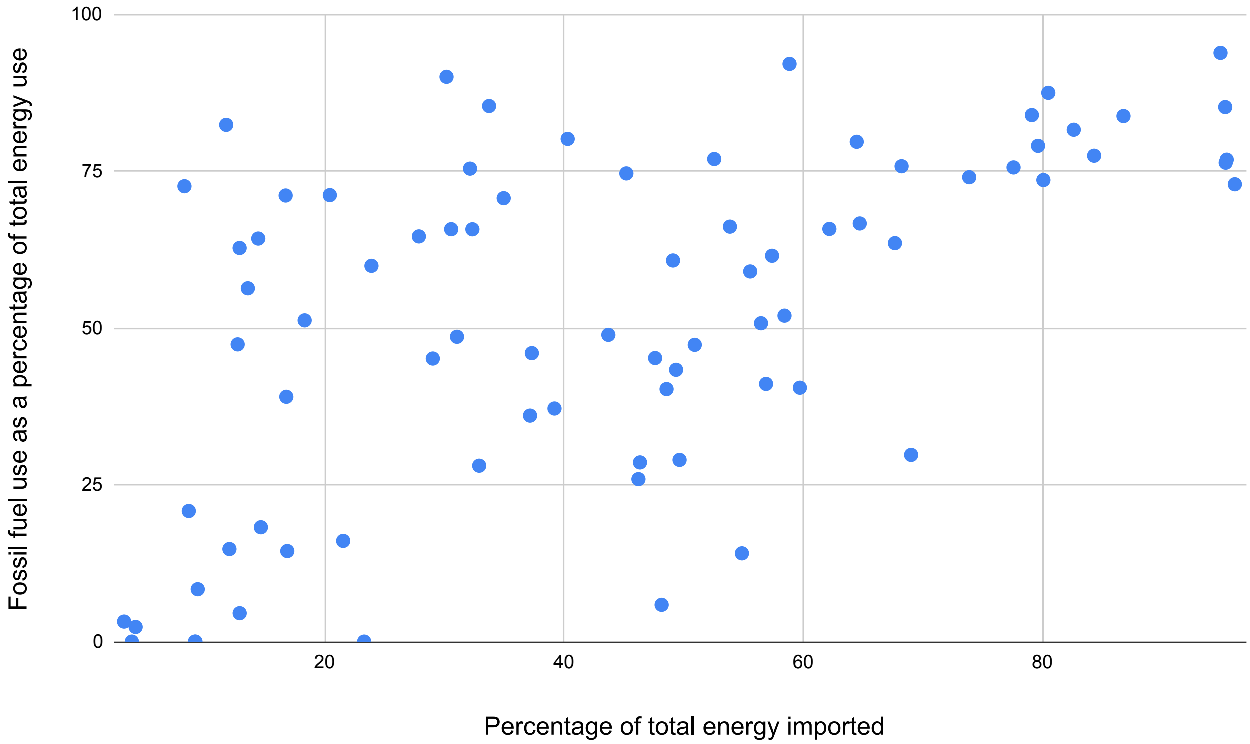

Figure 2: Scatter plot to show correlation between percentage of total energy imported against percentage of total energy derived via fossil fuels, per country. [7]
[8]

Energy Source	2000	2010	2020	2024
Total	-17%	30%	28%	44%

Table 1: UK fossil fuel import-export balance, negative meaning exports greater than imports.
[6]

Sector	1990	2000	2010	2020	2024
Industry	38.7	35.5	23.9	21.3	19.5
Domestic	40.8	46.9	48.3	39.4	34.0
Transport	48.6	55.5	55.3	41.7	54.0
Services	19.2	21.5	23.2	20.1	20.0

Table 2: UK inland energy consumption per year, measured in million tonnes of oil equivalent.
[6]

Energy Source	1990	2000	2010	2020	2024
Coal	229.9	120.0	107.6	5.7	2.0
Oil & other fuels	20.7	13.6	10.5	8.8	11.9
Gas	0.4	148.1	175.7	111.9	86.7
Nuclear	63.2	85.1	62.1	50.2	40.6
Hydro	5.6	5.1	3.6	6.9	5.8
Wind & Solar	–	0.9	10.3	88.2	97.7
Other renewables	–	4.3	12.3	38.6	40.3
Total	319.7	377.1	382.0	310.3	285.0

Table 3: UK electricity generated by fuel type, per year. (TWh) [6]

Energy Source	2000	2010	2020	2024
Onshore wind	0.9	7.2	34.9	34.7
Offshore wind	–	3.1	40.8	48.5
Solar PV	–	0.0	12.5	14.4
Hydro	5.1	3.6	6.9	5.8
Landfill Gas	2.2	5.2	3.5	2.9
Other Bioenergy	1.7	7.0	35.1	37.4
Total Renewables	9.9	26.2	133.6	143.7

Table 4: Electricity generation from renewables, per year (TWh) [6]

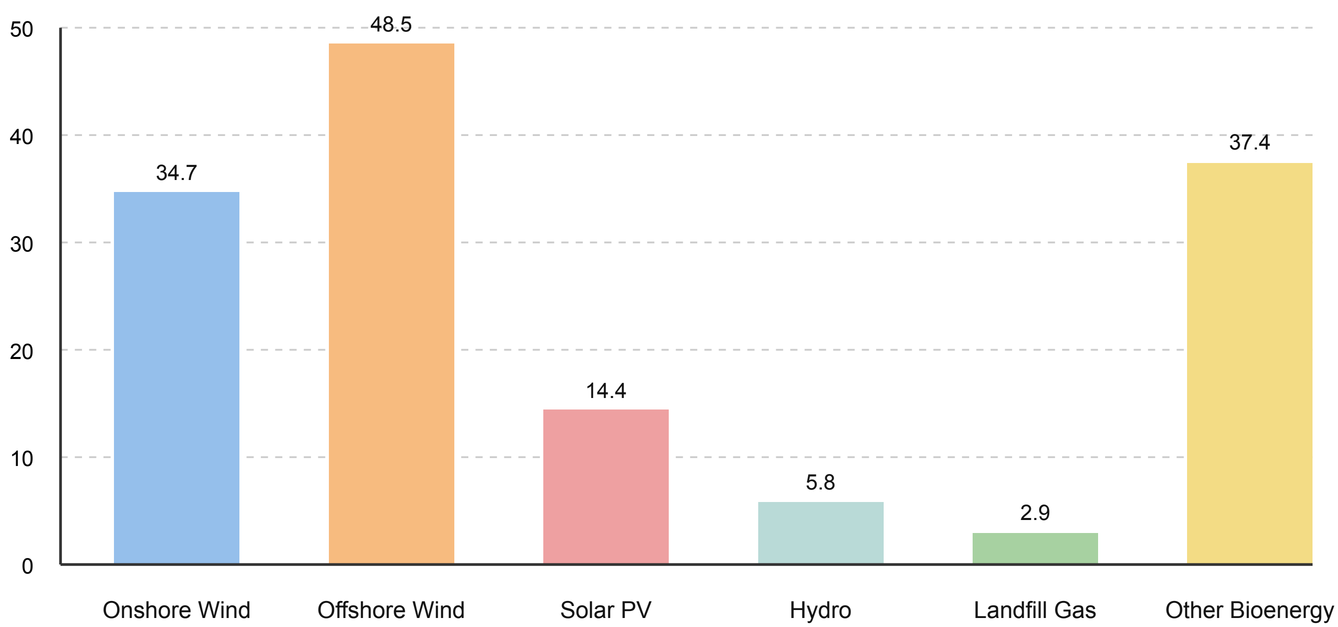

Figure 3: Electricity generation from renewables, 2024 (TWh)

Source	1990	2000	2010	2020	2022	2023	2024
Deep mined	72.9	17.2	7.4	0.1	0.1	0.1	0.1
Surface mining	19.9	14.0	11.0	1.6	0.6	0.4	0.0
Total Production	92.8	31.2	18.4	1.7	0.7	0.5	0.1
Coal imports	14.8	23.4	26.5	4.5	6.4	3.5	1.8

Table 5: UK coal mining and imports per year (million tonnes)
[6]

Sector	1990	2000	2010	2020	2024
Electricity generators	6.5	324.6	377.1	232.8	179.3
Energy Industries	39.2	102.1	95.9	88.7	72.5
Industry	164.6	198.5	118.0	100.7	85.4
Domestic	300.4	369.9	389.6	300.2	253.0
Services	86.4	110.5	101.6	89.4	92.0
Transport	–	–	–	0.9	1.8
Total	597.0	1105.5	1082.2	812.6	683.9

Table 6: UK natural gas usage per year (TWh)
[6]

Generation Type	1996	2000	2010	2020	2024
Conventional Steam	43.0	36.8	36.3	11.0	6.9
CCGT	12.7	22.9	34.1	31.6	31.3
Nuclear	12.9	12.5	10.9	7.8	5.9
Pumped Storage	2.8	2.8	2.7	2.7	2.7
Renewable	2.3	3.0	9.3	48.1	60.6
Total	73.6	77.9	93.2	101.3	107.5

Table 7: UK maximum electricity capacity (GW). Note that capacity usually significantly differs to production.
[6]

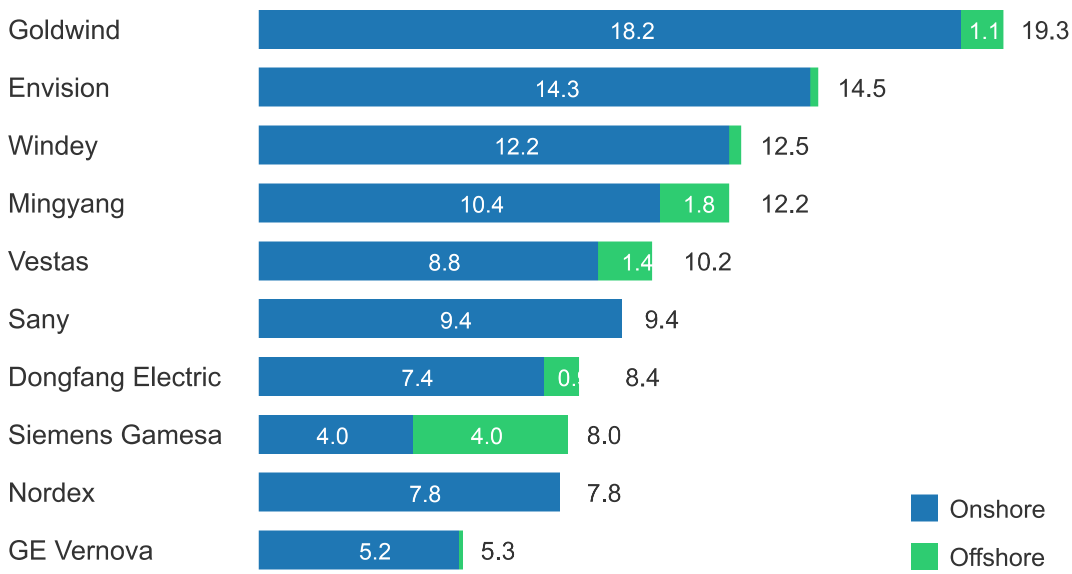

Figure 4: Bar chart to show GW capacity of wind turbines manufactured, per company, in 2024.[9]

Wind turbine manufacturing is increasingly dominated by Chinese private enterprises, primarily focused on onshore turbines. However, Siemens-Gamesa dominates offshore turbine manufacturing. Main offshore competitors include:

- Vestas of Denmark, began manufacture in 1979[10]
- Ming Yang of China

Ming Yang is developing floating wind turbines, using stabilising bases and anchors to operate in deeper waters. Siemens first trialed these concepts in 2009.[11]

Value is added through increasing energy generation capacity, minimising maintenance costs, maximising lifespan, improving transportability, and providing energy independence.

Part A.2 - Financial analysis

The following line charts regard data for the conglomerates as a whole (multinationals in the case of Vestas and Siemens Energy, and including stock investments and energy resale profits in the case of Ming Yang), which were extracted from public reports released by each company (non-trivial); see Appendix for raw data.

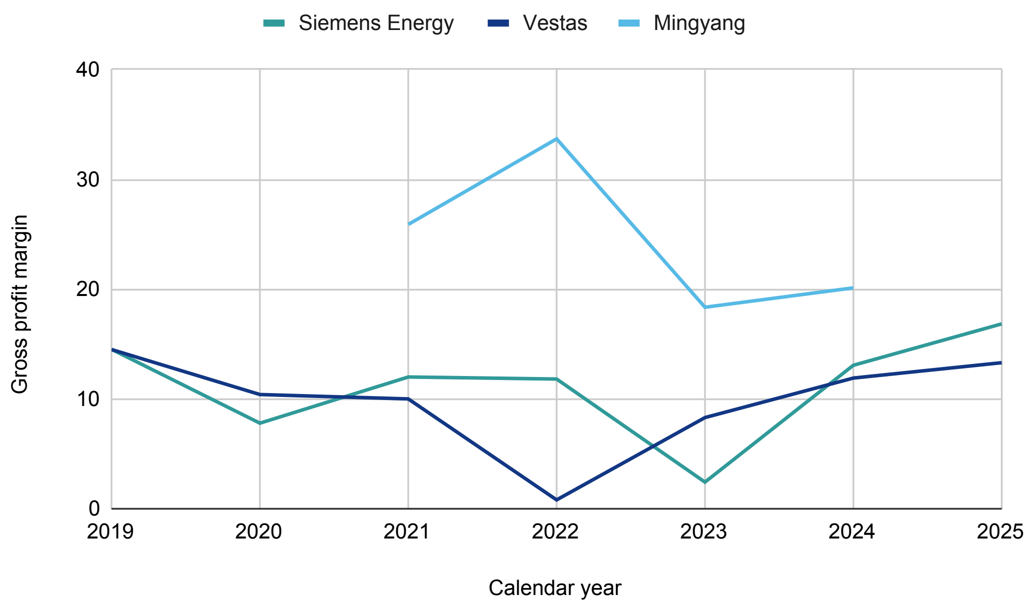

Figure 5: Line chart to show gross profit margin for 3 energy companies, where data available.[12][13][14]

Ming Yang's sales plateaued during 2022 and the invasion of Ukraine, which occurred 24th February 2022, and led to an increase in fossil fuel prices due to a boycott of importing fossil fuels from Russia. During this time period the total sales and assets of the company significantly increased, and greater government subsidies followed.

	2022	2023	2024
Domestic	4,362	2,987	1,969
Overseas	152	23	69

Table 8: Ming Yang gross profit at end of calendar year (in millions of CNY)[14]

	2021	2022	2023	2024
Ming Yang	61,550	68,940	83,861	86,795
Siemens Energy	44,141	51,084	47,907	50,874
Vestas	19,712	20,090	22,514	24,644

Table 9: Total assets at end of calendar year (in billions of CNY, millions of EUR).[12][13][14] Note that ¥1B CNY ≈ €123M EUR as of February 2026.

	2021	2022	2023	2024
Government subsidies	296	288	417	363

Table 10: Government subsidies made to Ming Yang per calendar year (in millions of CNY)[14]

The rise in the cost of fossil fuels led to significant increases in inflation, for any country that imports energy, or any country that imports from energy exporters.

Period	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**	**2025**	**2026**
Inflation Rate (%)	1.80	1.80	0.70	5.50	10.10	4.00	3.00	3.00

Table 11: Inflation rates, UK, for January of each year.[15]

Meanwhile countries besides Russia that are energy exporters saw greater demand from energy importers.

Resource	**2022**	**2023**	**2024**
Coal	142,345	103,188	85,154
Natural gas	90,343	74,322	67,441
Crude petroleum	14,424	11,467	10,649
Refined petroleum	4,241	4,323	5,023

Table 12: Largest energy exports of Australia (in millions of AUD)[16]

Country	**Growth within year**	**Total amount**
China	+13.2%	203,498
Japan	+23.7%	114,969
Korea, Republic of (South)	+2.0%	49,546
United States of America	+20.3%	32,556
India	-2.9%	32,405

Table 13: Largest importers of goods from Australia, 2022-2023 (in millions of AUD), growth in respect to previous year[17]

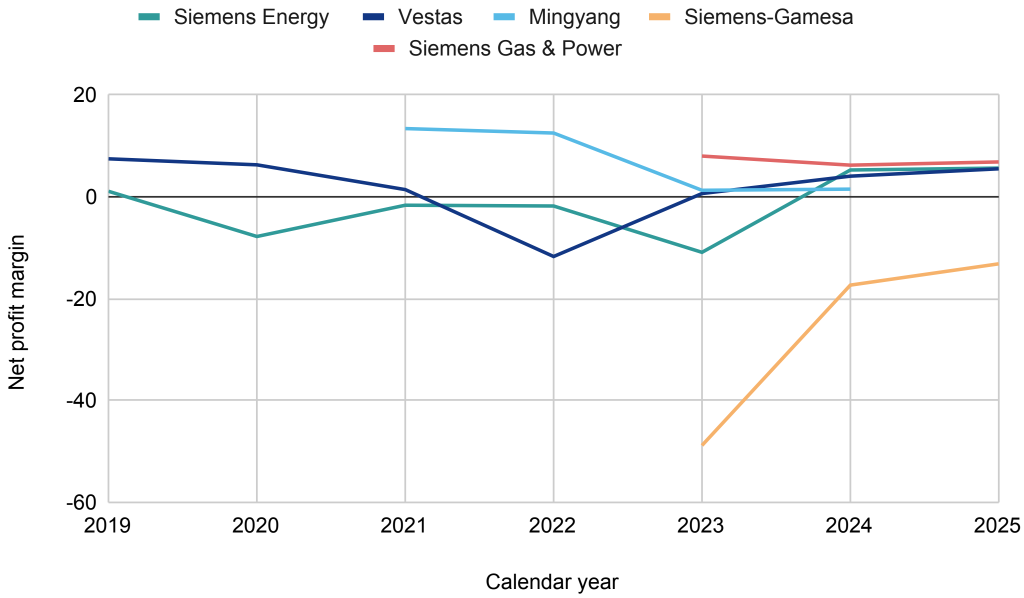

Figure 6: Line chart to show net profit margin for 3 energy companies, including 2 subsidiaries of 1 of the 3, where data available.[12][13][14]

Siemens Energy began to publicly breakdown net profit by subsidiary, which showed that Siemens-Gamesa saw growth in viability, but continued to be propped up by other subsidiaries. Ming Yang's net profits thinned out as boycotts of Russian fuel led to new exporters, for lower prices.

Overall it appears that wind turbine companies have a relatively thin net profit margin, despite barriers to entry in terms of production, transportation.

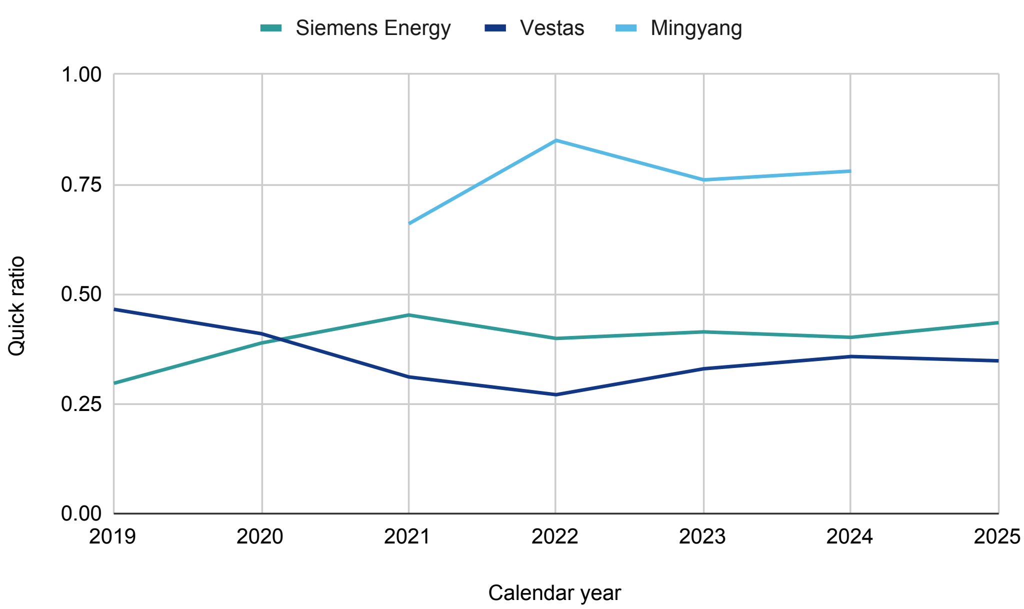

Figure 7: Line chart to show quick ratio for 3 energy companies, where data available.[12][13][14]

Based on the quick ratios, Ming Yang appears to be a more cash-oriented company, and the conglomerate holds many investments in other companies.

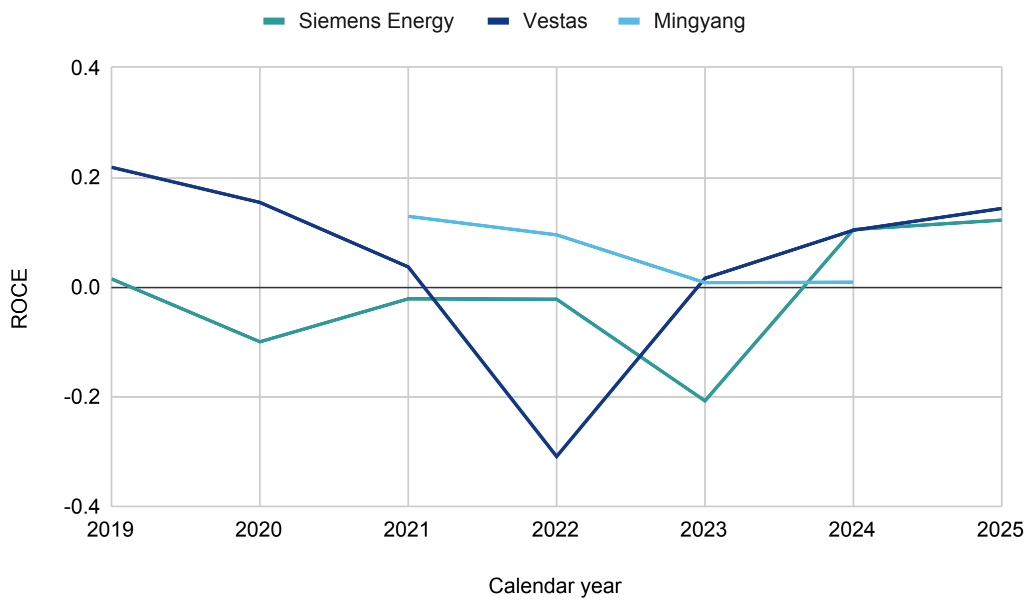

Figure 8: Line chart to show ROCE for 3 energy companies, where data available.[12][13][14]

The ROCE data suggests that wind turbine manufacturing generally is not a subsector within the energy sector which requires low capital employed, when comparing Ming Yang and Vestas, which solely produce wind turbines, to Siemens Energy as a conglomerate which handles other energy sources.

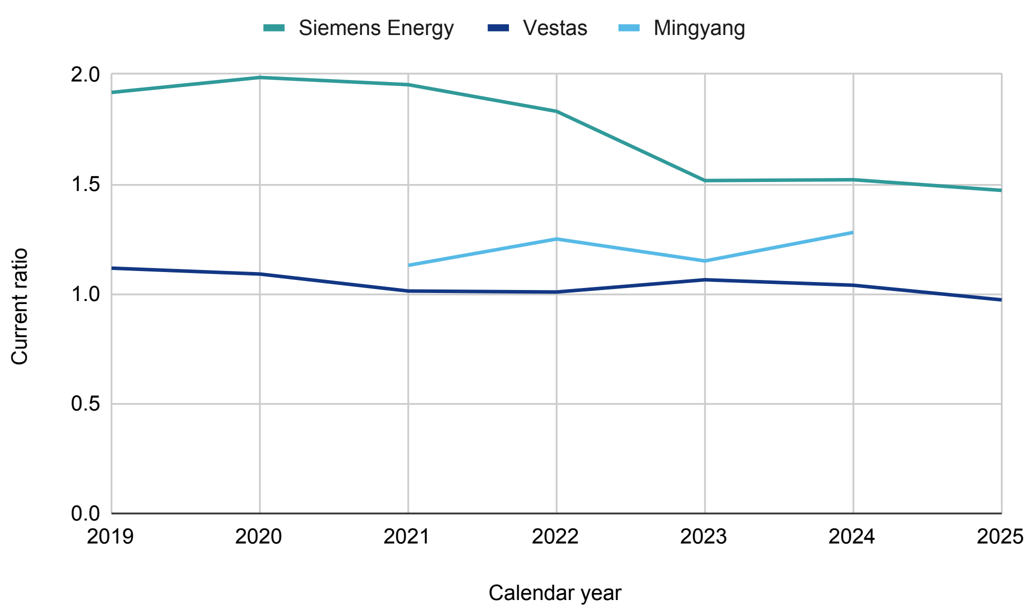

Figure 9: Line chart to show current ratio for 3 energy companies, where data available.[12][13][14]

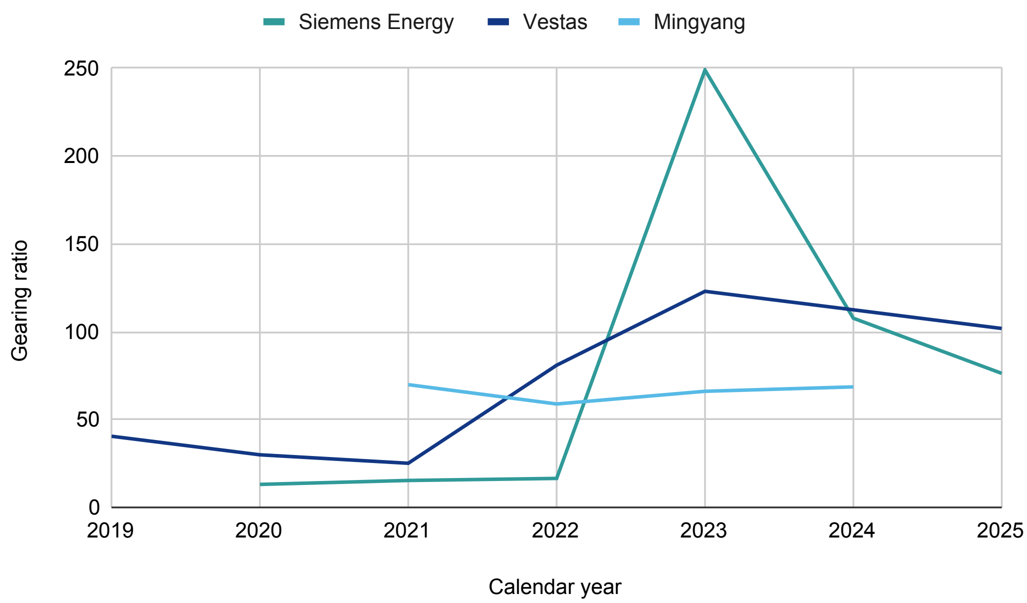

Figure 10: Line chart to show gearing ratio for 3 energy companies, where data available.[12][13][14]

The current ratio of Siemens Energy took a dip 2022, and remained lower. This means either assets dropped, or liabilities increased, the former seeming more plausible given the context. As some of Siemens Energy's products are oriented around increasing the efficiency of fossil fuel burning, it is possible this kind of easy-win equipment met significant demand, abruptly. The current ratio of Ming Yang, by contrast, increased, and this could be due to requiring more upfront payment, being a more cash-heavy company.

As many countries depend on fuel imports, it seems likely that increases in fossil fuel prices may lead to greater investments in renewable energy sources, for purposes of finances and energy independence. Over the past few decades, a sequence of events including the 2008 recession, US movements into extraction self-sufficiency, Covid-19, the invasion of Ukraine, and, recently, hostilities from Iran in the Middle East, have caused fluctuations in oil price, and fluctuations will likely continue.

Part A.3 - Change to existing product

A significant change to Siemens-Gamesa, or its parent conglomerate, would be to *copy Ming Yang*, and begin to invest in, and take a profit from, the sale of the energy generated by wind turbines, perhaps over a longer term contract.

China is one of the largest steel producers in the world.[18] Furthermore, steel typically makes up 66%-79% of a wind turbine.[19] Many Chinese steel companies are state-owned, and have been accused, collectively, of dumping excess quantities on the global market in pursuit of greater market share.[20] Furthermore, China is presently heavily impartial to the burning of coal for energy, a lower cost but highly polluting energy source. Estimates exist of steel production accounting for 20% of industrial energy use worldwide[21], indicating heavy coal use.

Region	Oil	Nat. Gas	Coal	Nuclear	Hydro	Renew.	Other	Total
China	9.3	297.8	5752.7	434.7	1226.0	1668.2	67.7	9456.4
Total Europe	50.5	644.7	523.0	735.6	642.6	1138.1	83.8	3818.3

Table 14: Energy generation by source (TWh), for Europe as a whole, and China.[22]

Labour costs are also lower in China than Europe. As a result, the overall costs of producing steel, and manufacturing products from steel, are significantly lower in China, and therefore it is difficult for European companies to compete solely in the manufacturing of wind turbines. However, simultaneously, as environmental standards are higher in Europe,[23][24] energy costs are generally higher. This means that every KWh produced for consumption within Europe has greater resale value than if it were produced for consumption within China.

To deduce how the costs of setup and electricity generation balance, the following facts are to be considered:

- Siemens-Gamesa recently did a deal with ScottishPower to supply 64x 14-236 DD wind turbines, with a total maximum generation capacity of 960MW, at a cost of £4billion[2]
- The East Anglia wind farm is expected to last a maximum of 25 years after construction[3]
- Total UK offshore wind power capacity is allegedly 10GW as of 2024[25]
- Generation from offshore wind was considered to be 48.5 TWh in 2024[6]
- The wholesale price according to supplier Octopus was, on average, £0.2077 per KWh for July 2024, for southern Scotland[26]
- The wholesale price according to Octopus was, on average, £0.2167 per KWh for July 2024, for London[26]
- The retail price, excluding 5% VAT, according to Octopus was £0.297 per KWh for July 2024, for London

We can then inspect how these facts interrelate.

$$\begin{aligned} 10_{GW} \cdot 24 \cdot 365 &= 87600\ GWh \\ \frac{87600}{10^3} &= 87.6\ TWh \\ \frac{48.5}{87.6} &= 0.5537 \\ 960_{MW} \cdot 24 \cdot 365 &= 8,409,600\ MWh \\ 8409600 \cdot 10^3 &= 8,409,600,000\ KWh \\ 8409600000 \cdot 0.5537 \cdot 0.2077 &= £967,133,350 \end{aligned}$$

So, the interpretation of the facts is that a modern wind farm, at a cost of £4b, will produce approximately £967.1m worth of electricity per year, at wholesale prices. 0.5537 represents the average UK ratio of how maximum capacity (if the wind was always blowing at an optimal speed

& direction) compares to actual electricity generation (note that it is not clear how maintenance costs relate to capacity threshold usage).

$$\frac{0.297}{0.2167} = 1.3706$$
$$1.3706 \cdot 967133350 = \pounds 1,325,512,713$$
$$1325512713 - 967133350 = \pounds 358,379,364$$
$$\frac{4000000000}{358379364} = 11.16$$
$$(25 - 11.16) \cdot \pounds 358,379,364 = \pounds 1,075,138,092$$

By considering the ratio between wholesale and retail prices where available, 1.3706, the retail markup of the energy generated by the new wind farm can then be ascertained. Ultimately, the new wind farm may viably produce enough energy to pay itself off in 11.2 years, using retail markup alone, based on 2024 energy prices (which may be influenced by global oil prices, which were at a relative midpoint, historically). Furthermore, an additional £1b profit is expected to be made, based on retail markup. If Siemens-Gamesa were then to take equity in this energy production, as part of the sale/setup, it looks like there is a long-term profit to be made, which could ultimately snowball into greater wind turbine production and larger profits.

The UK is a net importer of energy, particularly oil, chiefly used for transport fuels[27], which is frequently subject to geopolitical turbulence, including in the Strait of Hormuz. Furthermore, the adoption of electric cars is growing, nowadays more popular among new car purchases than diesel cars[28]. This means energy demands, and therefore prices, may increase. For contrast, France has comparatively cheap nuclear-powered electricity, retail prices being approximately that of the UK, at €0.194 (£0.17) per kWh[29].
Continuing, Ming Yang makes a significant proportion of its income from various investments in other energy companies, for example the world's largest hydroelectric dam, Three Gorges Dam, others of which could be resellers of energy produced by their own turbines.

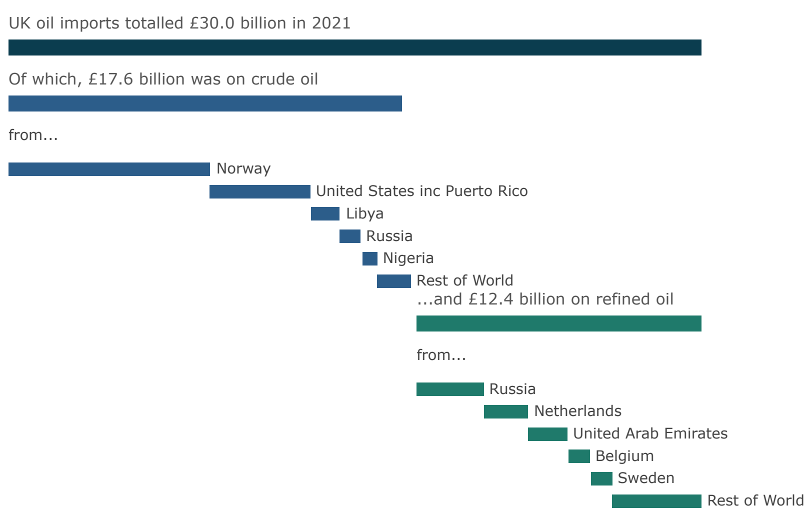

Figure 11: Bar chart to show UK oil imports, per country, 2021[30]

Investee	Opening Balance
Guangdong Yuecai Financial Leasing Co., Ltd.	439,621,649.49
Huaneng Mingyang New Energy Investment Co., Ltd.	3,327,573.18
Inner Mongolia Mingyang Northern Smart Energy Research Institute	60,678.13
Wuxi Mingyang Hydrogen Fuel Power Technology Co., Ltd.	8,043,192.69
Golmud Mingyang New Energy Power Generation Co., Ltd.	33,846,380.38
Panzhihua Renhe Jieyuan New Energy Co., Ltd.	2,437,535.47
Chengde County Shantaijieyuan Steel Structure Co., Ltd.	18,866,128.76
Three Gorges New Energy (Fenghuang) Power Generation Co., Ltd.	45,600,000.00
GCPC Investors Limited	5,405,420.17
CSC Mingyang New Energy REITs Fund	247,342,004.02
Shuimu Mingtuo (Damao) Energy Management Co., Ltd.	212,597.30
Guangxi Guangtou Beibu Gulf Offshore Wind Power Co., Ltd.	37,364,051.32
Hainan Prefecture Jinyuan Qieji Wind Power Co., Ltd.	10,862,670.95
Inner Mongolia Eastern Electric Power Trading Center Co., Ltd.	5,432,895.36
Sichuan Xingong Green Hydrogen Technology Co., Ltd.	–
China Resources New Energy (Yangjiang Yangdong) Co., Ltd.	3,191,500.00
Guohua (Shanwei) Wind Power Co., Ltd.	2,680,786.90
Mengxi New Energy Development (Baotou) Co., Ltd.	1,000,000.00
China Resources Wind Power (Shanwei) Co., Ltd.	833,100.00
Datang Baoshan New Energy Co., Ltd.	338,000.00
Shanwei Huadian Energy Co., Ltd.	260,000.00
Southern Offshore Wind Power Joint Development Co., Ltd.	70,000,000.00

Table 15: Long-term equity and other equity instrument investments of Ming Yang Smart Energy Group, 2025[14] (CNY)

Item	**H1 2024**	**H1 2025**
Investment income	113,767,682.13	214,306,799.41
Including: Gains from investment in associates and joint ventures	9,660,544.75	14,486,443.27
Total comprehensive income	686,687,823.50	632,381,770.52
Investment percentage of total income	17.97%	36.18%

Table 16: Investment income and total income comparison, for first half of years, of Ming Yang Smart Energy Group[14] (CNY)

SSE, a former Scottish state monopoly now publicly traded company, is currently involved in developing the largest offshore wind farm in the UK, Dogger Bank Phases A, B, C, D[31][32], involving GE wind turbines. SSE's current total renewables fleet has a capacity of 5GW (1.5GW hydro, 2.5GW onshore), and SSE are aiming to grow offshore wind significantly[33]. The stock price, listed LSE:SSE, has grown from £1,170 per share in 2010 (£1,831 when adjusted for inflation, up to 2026) to £2,583 in 2026[34]. If Ming Yang were British, they would potentially invest in this company.

RWE is involved in physically adjacent projects[35], named Sofia and Dogger Bank South, the former involving 100 Siemens-Gamesa turbines[36] and 100% RWE ownership, the latter of 51% ownership[32]. RWE's stock price has gone from a high of €95.6 in 2007, to a low of €10.15 in 2015, and is currently on a high of €58.3[37]. This is potentially another kind of investment Ming Yang would make.

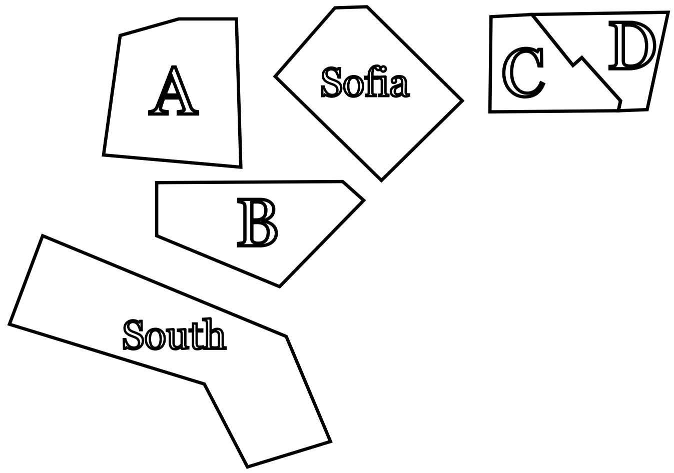

Figure 12: Geographical layout of Dogger Bank offshore wind farm projects.[7]

Project	**Status**	**Gross Capacity**	**SSE Ownership**
Greater Gabbard	Operational	0.5GW	50%
Beatrice	Operational	0.6GW	40%
Seagreen 1 & 1A	Operational / In Development	1.1GW / 0.36GW	49%
Dogger Bank A, B, C	In Construction	3.6GW	40%
Dogger Bank D	In Development	1.5GW	50%
IJmuiden Vers Alpha	In Development	2GW	50%
North Falls	In Development	0.5GW	50%
Ossian	In Development	3.6GW	40%

Table 17: SSE UK offshore wind project portfolio and ownership details[33]

Project Name	**Status**	**Gross Capacity**
Robin Rigg	In Operation	0.17GW
Gwynt y Môr	In Operation	0.58GW
Rhyl Flats	In Operation	0.0GW9
Triton Knoll	In Operation	0.86GW
Humber Gateway	In Operation	0.22GW
Scroby Sands	In Operation	0.06GW
Greater Gabbard	In Operation	0.50GW
Galloper	In Operation	0.35GW
Rampion	In Operation	0.40GW
London Array	In Operation	0.63GW
Sofia	Under Construction	1.40GW
Awel y Môr	In Development	0.58GW
Dogger Bank South (West)	In Development	1.50GW
Dogger Bank South (East)	In Development	1.50GW
Vanguard (West)	In Development	1.40GW
Boreas	In Development	1.40GW
Vanguard (East)	In Development	1.40GW
Five Estuaries	In Development	0.35GW
North Falls	In Development	0.50GW
Rampion 2	In Development	1.20GW

Table 18: RWE UK offshore wind project portfolio[38]

Part B.1 - Risks

Risk	Risk type	Severity	Example
Fault in functionality	Financial risk	Yellow	1980 - Vestas' production stopped, compensation paid to customers for a year due to fault[39].
Cash issues	Financial risk	Yellow	2023 - Siemens Energy suffered liquidity issues, and needed government support to fulfill orders[40].
Transport issues	Operational risk	Yellow	2006 - a giant jack-up barge collided with an offshore wind turbine in the Scroby Sands wind farm[41].
Geographical issues	Operational risk	Green	While modern monopiles can allow wind turbines to be installed at depths of 60m, initial layers of the seabed may be very soft and of low density[42].
Skilled labour issues	Operational risk	Yellow	UK head of Siemens Energy has talked of the biggest challenge facing growth being a skilled workforce[43].
Occupational hazards	Hazard risk	Green	2005 - worker fell approximately 200 feet to the ground, struck an electrical transformer box, and was killed[44].

Table 19: Risks associated with wind turbine manufacturing, part 1

Risk	Risk type	Severity	Example
Geopolitical issues	Strategic risk	Red	2024 - Finland-Estonia undersea electricity cable cut, suspected Russian ship[45].
Maintenance issues	Strategic risk	Green	2025 - 15-tonne blade falls from turbine [46]
Weather issues	Financial risk	Green	2022 - high winds caused a £20 million wind turbine to collapse, damaging blades[47].
Public opinion	Strategic risk	Green	2025 - representative survey finds opposition to wind farms due to noise, visuals, wildlife issues[48].
Supply chain environmental issues	Hazard risk	Yellow	Metallurgical coal is emissions heavy, typically used in blast furnaces[49].
Oil abundance	Strategic risk	Red	2020 - crude oil is traded for $9.12 per barrel[50].
Pandemic	Financial risk	Yellow	2022 - inflation rate reaches 11% [51].

Table 20: Risks associated with wind turbine manufacturing, part 2

The hypothetical premises that Siemens-Gamesa may setup in Manchester could be an R&D facility, given the student population, abundance of equipment, and dominant physics culture.

The R&D facility seeks to find the best solutions to common turbine faults, in tandem with product development. Issues regarding standardisation of metrics do exist[52].

Probability

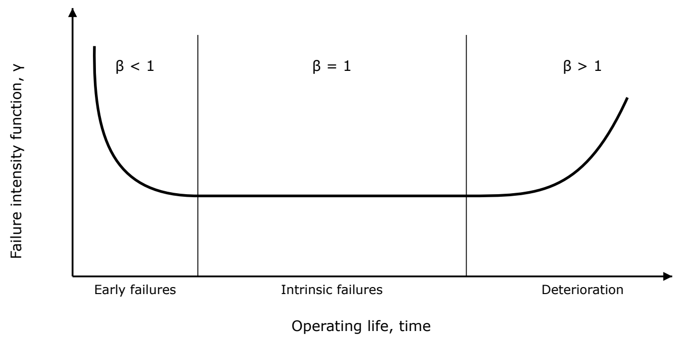

Figure 13: Failure intensity over product lifespan[52].

In the case of 11 failure types are recorded, for which 0 or more can occur in a year, the total possible combinations of failures is:

$$\sum_{i=0}^{N} {}^{N}C_i$$

If each failure type is assumed to be independent, the probabilities within each combination can be multiplied:

Axiom 1 (Product of independent events)

$$P(A \cap B) = P(A) \cdot P(B)$$

A viable initial set of probabilities of failure rates per year, for different fault types, is[52]:

$$R = \{0.55, 0.42, 0.25, 0.24, 0.18, 0.17, 0.14, 0.12, 0.11, 0.11, 0.06\}$$

However, these each exist within independent probability spaces.

Axiom 2 (Quantification of full probability space)

$$P(\Omega) = 1$$

And so the probabilities are then normalised via $\frac{p_i}{\sum_i p_i}$ as follows:

$$S = \{0.2340, 0.1787, 0.1064, 0.1021, 0.0766, 0.0723, 0.0596, 0.0511, 0.0468, 0.0468, 0.0255\}$$

The algorithm to calculate the overall failure probability is then:

1. Generate all possible combinations
2. Iterate through all possible combinations, calculating product for each
3. Add all products together

The probability that any possible combination of faults occur within 1 year, per turbine, is then:

$$p = 0.6906$$

Calculations at:

DOI 10.5281/zenodo.19741763

Note that the probability that a maintenance issue will occur is considered to depend upon the lifetime of the turbine, as in Figure 13; turbines most often exhibit faults when new, or with old age[52].

Potential impact

#	Category	Failure rate
1	Pitch / Hyd	1.080
2	Other Components	1.010
3	Generator	1.000
4	Gearbox	0.640
5	Blades	0.520
6	Grease / Oil / Cooling Liq.	0.475
7	Electrical Components	0.440
8	Contactor / Circuit Breaker / Relay	0.435
9	Controls	0.430
10	Safety	0.400
11	Sensors	0.350
12	Pumps / Motors	0.350
13	Hub	0.240
14	Heaters / Coolers	0.220
15	Yaw System	0.190
16	Tower / Foundation	0.185
17	Power Supply / Converter	0.180
18	Service Items	0.130
19	Transformer	0.070

Table 21: Component failure rate per offshore turbine, per year, from a study involving wind turbines of age 3+[53].

Category	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Major Replacement	0.001	0.001	0.095	0.154	0.001	0.000	0.002	0.002	0.001	0.000	0.000	0.000	0.001	0.000	0.001	0.000	0.005	0.000	0.001
Major Repair	0.179	0.042	0.321	0.038	0.010	0.006	0.016	0.054	0.054	0.004	0.070	0.043	0.038	0.007	0.006	0.089	0.081	0.001	0.003
Minor Repair	0.824	0.812	0.485	0.395	0.456	0.407	0.358	0.326	0.355	0.373	0.247	0.278	0.182	0.190	0.162	0.092	0.076	0.108	0.052
No Cost Data	0.072	0.150	0.098	0.046	0.053	0.058	0.059	0.048	0.018	0.015	0.029	0.025	0.014	0.016	0.020	0.004	0.018	0.016	0.009

Table 22: Ratio of costs of repair/replacement, where minor repair is <€1000, major repair is €1000-€10,000, major replacement is €10,000+, from a study involving wind turbines of age 3+[53].

A rough estimate of the yearly costs can be made by assigning:

- A unique value per category failure rate, f_i, for F categories
- A cost to each of repair/replacement category, m_i, for N categories
- A ratio value to the severity of each replace/replacement, s_j, for N categories

This results in the following equation:

$$\sum_{i=0}^{F}\sum_{j=0}^{N} f_i m_j s_j$$

The component-specific failure rates can be redacted as follows:

$$\sum_{j=0}^{N} m_j s_j$$

In the case that the costs are $\{500, 5000, 15000\}$, the average cost, per turbine, per year is €61,097.20, or, without considering the probability of individual component failures, is €98,112.5.

When it is considered that a wind farm may contain 100 wind turbines; a maximum potential of €6m in maintenance costs per year in this case. Careful planning needs to be executed to guarantee reliability of electricity delivered, and efficiency in the setup and connection, as costs are ongoing, as error may result in lost investments, which are large upfront. Eventual owners of wind farms will need to maintain binding contracts with turbine manufacturers, and select models carefully, as a result.

Overall rating

The EMV is then, per year, for a wind farm of 100 turbines...

When considering the cost ratios provided by Carroll and the failure rates by Zappala et al.[52]:

$98110 \cdot 0.6906 \cdot 100 = €6,776,650$

When considering data solely provided by Carroll et al.[53]:

$61100 \cdot 100 = €6,110,000$

Both relatively similar, which inspires confidence in the failure rate probabilities.

Potential solutions and strategies

Wind turbines contain sensors - torque, vibration, sound, temperature, etc. - which produce data, the data then used to deduce the state of the turbine, e.g. by running the data through neural networks to detect interrelations, and therefore malfunctions, including of individual sensors themselves[54]. Sensor systems, known as SCADA, are common across manufacturers.

The maintenance schedule can be considered an optimisation problem, containing variables such as downtime, cost of maintenance tasks, cost of generation disruption, etc. This can then be solved via algorithms such as the Genetic Algorithm[55].

Modern SCADA products may regularly upload data to a data centre, where it is pipelined through automated processing for immediate end-user presentation of findings[56]. Predictive weather data can allow for power generation forecasting[57].

For the above strategies, costs involve skilled labour to inspect and handle instructions for the data; the numerical processing and storing itself is inexpensive. Viably, a tech company may be subcontracted for setup, £400,000, and intermittent upkeep performed, £100,000 annually.

More expensively, retrofitting older models with newer, more fault-tolerant components is also viable. Prices for parts not publicly available, $62,580 salary for a technician.[58]

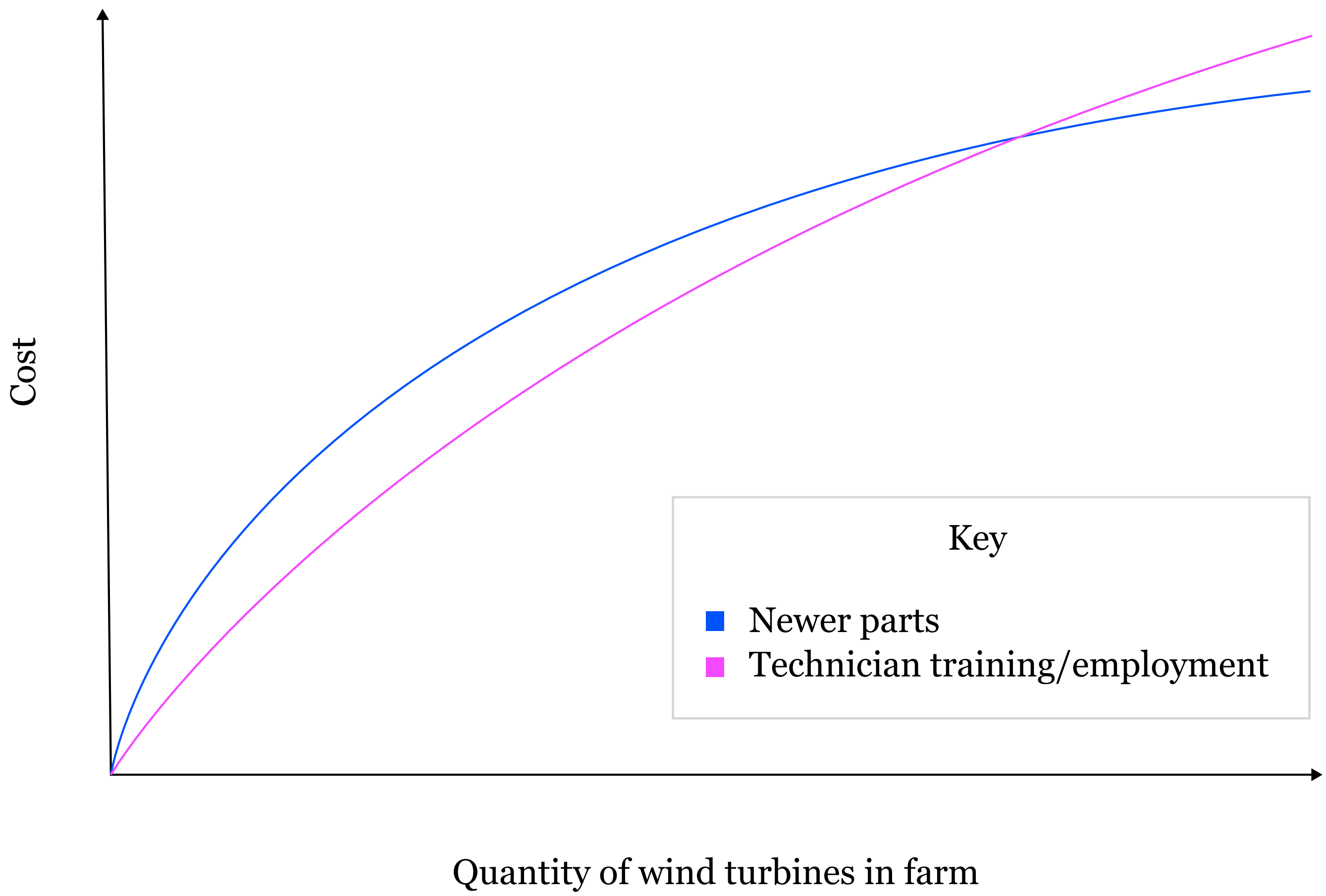

Figure 14: Example line chart (exact numerical values unavailable) to demonstrate how retrofitting of wind turbines with newer parts, pre-emptively, may compare to maintenance costs of only reacting to parts that develop faults, as a wind farm is scaled.

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Appendix

Energy Usage and Import Statistics (related to scatter graph in Part A.1)

Country Name	Imports (%)	Fossil Fuel Use (%)
Albania	12.70	47.35
Argentina	13.55	56.29
Armenia	67.63	63.48
Austria	62.15	65.74
Bangladesh	20.40	71.13
Belgium	95.27	76.30
Bosnia and Herzegovina	33.71	85.33
Botswana	34.93	70.64
Brazil	8.61	20.81
Bulgaria	37.13	36.00
Chile	68.18	75.73
China	16.71	71.07
Costa Rica	50.90	47.28
Cote d'Ivoire	3.21	3.19
Croatia	49.63	28.95
Cuba	54.84	14.06
Czechia	32.12	75.35
Denmark	14.42	64.21
Dominican Republic	95.36	76.78
Egypt, Arab Rep.	12.86	62.71
El Salvador	58.40	51.93
Estonia	12.01	14.75
Eswatini	23.27	0.00
Ethiopia	9.36	8.34
Finland	48.54	40.23
France	47.58	45.18
Georgia	73.83	73.98
Germany	64.44	79.63
Greece	79.07	83.88
Guatemala	39.18	37.14
Honduras	43.68	48.87
Hungary	53.84	66.11
India	37.28	45.96
Ireland	95.23	85.16
Italy	79.59	78.98

Country Name	Imports (%)	Fossil Fuel Use (%)
Japan	94.84	93.79
Kenya	21.52	16.04
Korea, Rep.	86.73	83.72
Kyrgyz Republic	64.69	66.62
Lao PDR	3.85	0.00
Latvia	55.53	58.97
Moldova	82.57	81.55
Montenegro	30.54	65.70
Morocco	96.04	72.86
Namibia	77.54	75.54
Nepal	16.84	14.43
Netherlands	58.82	92.04
New Zealand	23.88	59.87
Nicaragua	46.19	25.88
North Macedonia	52.52	76.88
Pakistan	29.00	45.10
Philippines	49.34	43.30
Poland	30.15	89.98
Portugal	84.28	77.41
Romania	16.76	39.01
Rwanda	9.13	0.00
Senegal	68.98	29.74
Serbia	27.84	64.55
Slovak Republic	57.36	61.47
Slovenia	49.08	60.72
Spain	80.04	73.53
Sri Lanka	56.86	41.05
Sudan	4.17	2.33
Sweden	32.88	28.02
Switzerland	56.43	50.72
Syrian Arab Republic	59.68	40.45
Tajikistan	31.03	48.56
Tanzania	14.64	18.22
Thailand	46.32	28.55
Tunisia	45.18	74.59
Turkiye	80.44	87.43
Uganda	9.17	0.00
Ukraine	32.31	65.69
United Kingdom	40.28	80.09
United States	11.74	82.33
Uruguay	48.13	5.86
Viet Nam	8.25	72.55
Zambia	12.87	4.53
Zimbabwe	18.30	51.18

Siemens Energy

Financial data of Siemens Energy (Values in millions of EUR)

Category	2019	2020	2021	2022	2023	2024	2025
Profit before tax	317	-2135	-465	-518	-3387	1822	2213
Retained profit	11472	2906	2605	2453	-6583	-5578	-3990
Long-term debt	547	1672	2177	2474	3190	3287	3287
Current debt		718	551	749	1591	479	1528
Debtors	5097	3786	5110	5572	6537	7072	7571
Cash	1871	4630	5110	5572	6537	6363	9162
Revenue	28797	27457	28482	28997	31119	34465	39077
Gross profit	4181	2139	3417	3425	753	4503	6579
Shareholder capital	13089	15390	15220	17122	8503	9075	10301
Capital employed	21,554	21,363	21,539	23,232	16,308	17,403	18,146
Total current assets	45,041	43,032	44,141	51,173	47,907	50,874	56,637
Total current liabilities	23,487	21,669	22,602	27,941	31,599	33,471	38,491
Revenue (SEGP)			20888	26883	12897	16365	22996
Revenue (SG)	10226	9482	10198	9814	9092	10008	10375
Orders (SG)	12749	14736	12185	11598	16836	7256	9324
Profit before tax (SEGP)					1035	1021	1580
Profit before tax (SG)					-4439	-1734	-1364

- Metrics for each year were taken from that year's annual financial company report
- SG: Siemens-Gamesa
- SEGP: Siemens Energy gas and power

Vestas

Financial data of Vestas (Values in millions of EUR)

Category	2019	2020	2021	2022	2023	2024	2025
Cash and equivalents	2,888	3,063	2,420	2,378	3,318	3,817	4,384
Debtors	1,460	1,538	1,531	1,280	1,305	1,719	1476
Total current liabilities	9,347	11,232	12,700	13,513	14,020	15,480	16,847
Total current assets	10,442	12,247	12,864	13,628	14,921	16,089	16,386
Capital employed	4,165	6,057	6,133	5,487	6,429	6813	7255
Profit before tax	909	934	224	-1,696	102	705	1,039
Revenue	12147	14819	15587	14486	15382	17295	18822
Equity	3345	4703	4697	3060	3042	3542	3881
Total debt	1354	1407	1179	2476	3742	3984	3953

- Metrics for each year were taken from that year's annual financial company report
- Gross profit margin: Calculated using Gross profit and Revenue (used as turnover).
- Net profit margin: Calculated using Profit/(loss) before tax, and Revenue (or turnover).
- Gearing: Calculated using total financial debts and equity (gross gearing).
- ROCE: Calculated via turnover and capital employed (note that the company omits 'special items' in their own ROCE calculations, which scales all ROCE values towards 0 more so).

Ming Yang Smart Energy Group Co., Ltd.

Financial data of Ming Yang (Values in billions of CNY)

Metric	2021	2022	2023	2024
Turnover (operating revenue)	27.16B	30.75B	27.86B	27.16B
Operating costs (all)	21.34B	24.60B	24.74B	24.96B
Raw materials cost	19.66B	17.97B	20.99B	21.91B
Labour cost	0.39B	0.48B	0.63B	0.71B
Turnover (operating revenue)	27.16B	30.75B	27.86B	27.16B
Operating costs	21.34B	24.60B	24.74B	24.96B
Profit before tax	3.65B	3.86B	0.36B	0.41B
Total cash (Cash and bank balances)	14.07B	11.16B	13.04B	14.58B
Debtors (Accounts receivable)	5.86B	10.77B	14.14B	13.78B
Raw materials costs	19.70B	19.83B	22.05B	20.98B
Labour costs	0.42B	0.56B	0.70B	0.71B
Total assets	61.55B	68.94B	83.86B	86.80B
Total liabilities	43.00B	40.58B	55.41B	59.53B
Current liabilities	33.25B	28.36B	37.82B	39.14B
Capital employed	28.30B	40.58B	46.04B	47.66B
Current assets	37.69B	35.46B	43.73B	50.13B
Capital employed	28.30B	40.58B	46.04B	47.66B

- Metrics for each year were taken from that year's annual financial company report
- Ratios directly extracted from company financial report: quick ratio, current ratio, gearing ratio, however manual calculations for quick and current produced similar results.
- Net margin based on profit before tax sourced via 'Reconciliation between income tax expenses' tables
- Gross margin based on subtracting raw materials and labour cost metrics from full company turnover, via 'By Industry' table
- Total assets and total liabilities sourced via 'Changes in the total number of shares and the shareholder structure of the Company'